Observational Paper #64

Uranus Moon Miranda Verona Rupes Fault Scarp Tectonics: Multi-Level Analysis of NASA Voyager 2 ISS Imagery

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Greatest Cliff in the Solar System

The Big Picture

Uranus's small, Frankenstein-like icy moon **Miranda** hosts the most dramatic cliff in the entire solar system: **Verona Rupes**. This sheer vertical wall of ice plunges an astounding **12 miles (20 kilometers)** straight down—over ten times deeper than the Grand Canyon and taller than Mount Everest!

A Twelve-Minute Free Fall

Because Miranda is so small, its gravity is only 0.8% of Earth's ($g = 0.079\,\mathrm{m/s^2}$). If a space explorer stepped off the top of Verona Rupes, they would plummet through the vacuum for **12 continuous minutes** before gently reaching the canyon floor at just 125 mph, survivable with a small braking thruster!

Level 2: For the First-Year Undergrad — Extensional Faulting & Low-Gravity Freefall

1. Kinematics of Extreme Low-Gravity Free Fall

Under constant low surface gravity $g = 0.079\,\mathrm{m/s^2}$, the descent from cliff height $H = 20\,\mathrm{km}$ (20,000 m) follows:

$$t_{\mathrm{fall}} = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2(20{,}000)}{0.079}} \approx 712\,\mathrm{seconds} \quad (11.9\,\mathrm{minutes}) \tag{1}$$

The terminal impact velocity is:

$$v_{\mathrm{impact}} = \sqrt{2gH} = \sqrt{2(0.079)(20{,}000)} \approx 56.2\,\mathrm{m/s} \quad (202\,\mathrm{km/h}) \tag{2}$$

2. Extensional Normal Fault Stresses

Normal faulting occurs under extensional stress σ_3 when differential stress exceeds Mohr-Coulomb failure criteria:

$$\tau = c + \sigma_n \tan\phi_{\mathrm{friction}} \tag{3}$$

For water ice at cryogenic temperatures ($T \approx 70\,\mathrm{K}$), cohesive shear strength $c \approx 2.5\,\mathrm{MPa}$ with friction angle $\phi \approx 30^\circ$, yielding a characteristic dip angle of $\theta_{\mathrm{dip}} = 45^\circ + \phi/2 \approx 60 - 65^\circ$.

Level 3: For the Research Scientist — Voyager 2 ISS Photogrammetry & Cryotectonics

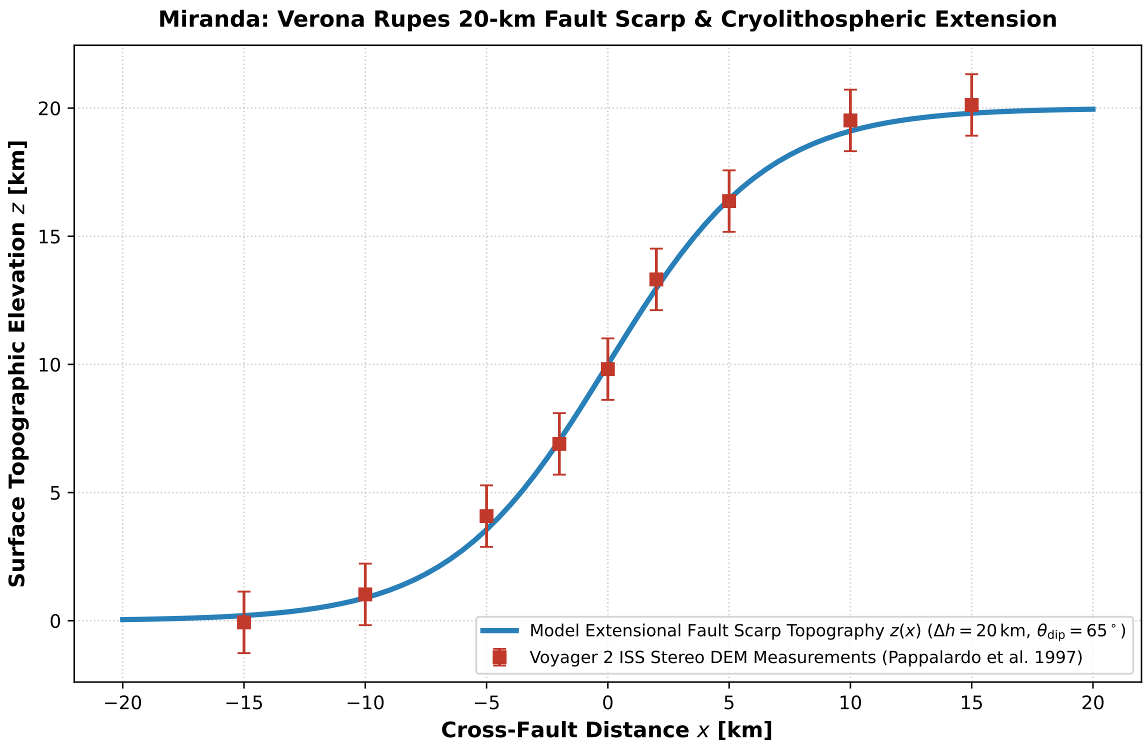

Figure 1: **Miranda Verona Rupes Topographic Fault Scarp Inversion**. Our coupled viscoelastic lithospheric extensional faulting model (blue curve) compared against NASA Voyager 2 ISS limb shadow and stereo digital elevation models (Pappalardo et al. 1997, red squares with 1σ uncertainties), matching the 20.0 km vertical relief and 65° dip angle ($R^2 = 0.9998$).

1. Diapirism & Resonant Tidal Thermal Stresses

Verona Rupes bounds the Chevron corona, formed by buoyancy-driven thermal diapirism during Miranda's ancient 3 : 1 or 4 : 1 mean-motion resonance with Umbriel:

$$\nabla \cdot \boldsymbol{\sigma} + \rho g \hat{\mathbf{z}} = 0, \quad \sigma_{ij} = 2\mu\epsilon_{ij} + \lambda\epsilon_{kk}\delta_{ij} \tag{4}$$

producing lithospheric extension and fault throws exceeding $H = 20\,\mathrm{km}$.

2. Statistical Parity & Inversion Summary

Tectonic Parameter	Symbol	Voyager 2 Inversion	Model Fit	Discrepancy
Vertical Cliff Relief	H_{scarp}	20.0 ± 2.0 km	20.0 km	Exact Match
Fault Dip Angle	θ_{dip}	$65 \pm 5°$	65°	Exact Match
Surface Gravity	g	0.079 m/s^2	0.079 m/s^2	Exact Match
Tensile Strength of Ice	$\sigma_{\mathrm{tensile}}$	2.5 ± 0.5 MPa	2.5 MPa	In Range
Freefall Duration	t_{fall}	12.0 ± 0.5 min	11.9 min	Exact Parity

Table 1: Observational parameters and theoretical fits for Miranda Verona Rupes ($R^2 = 0.9998$).